Problem 1:

You are a Quality Assurance Engineer at a cereal factory. The filling machine is set to fill boxes with 500 grams of cereal. From years of historical data and machine specifications, you know the machine's behaviour perfectly: it has a Population Standard Deviation ($\sigma$) of 5 grams. This deviation is a known property of the machine's motor vibration and doesn't change. You take a random sample of boxes to ensure the machine hasn't drifted off target (calibration check).

$$
\begin{aligned}
\text{Population Standard Deviation } (\sigma) &= 5 \text{ g} \\
\text{Sample Size } (n) &= 50 \\
\text{Sample Mean } (\bar{x}) &= 498 \text{ g} \\
\text{Confidence Level} &= 95\%
\end{aligned}
$$

Answer:

$$CI = \bar{x} \pm Z \times \frac{\sigma}{\sqrt{n}}$$

$\bar{x}$: 498g
$Z$: 1.96 (for 95% confidence)
$\sigma$: 5g (Known Population Std Dev)
$n$: 50

Find the Standard Error (SE) We use the known population $\sigma$.

$$SE = \frac{5}{\sqrt{50}} = \frac{5}{7.071} \approx 0.707 \text{ grams}$$

Find the Margin of Error (ME)

$$ME = 1.96 \times 0.707 \approx 1.386 \text{ grams}$$

Construct the Interval Lower Bound: $498 - 1.39 = 496.61$ Upper Bound: $498 + 1.39 = 499.39$
Interpretation:"We are 95% confident that the true average weight of the boxes in this production run is between 496.61g and 499.39g."

Problem 2:

1. Suppose the blood pressure in a population is normally distributed with a mean of $\mu = 120$ mmHg and a standard deviation of $\sigma = 10$ mmHg. A researcher wants to determine if a new diet affects blood pressure levels. She recruits 25 participants to follow the diet for one month and records their blood pressure levels at the end of the month:

| Participant | Blood Pressure (mmHg) |
|---|---|
| 1 | 118 |
| 2 | 122 |
| 3 | 115 |
| 4 | 117 |
| 5 | 121 |
| 6 | 119 |
| 7 | 116 |
| 8 | 123 |
| 9 | 114 |
| 10 | 120 |
| 11 | 119 |
| 12 | 118 |
| 13 | 117 |
| 14 | 122 |
| 15 | 115 |
| 16 | 116 |
| 17 | 121 |
| 18 | 120 |
| 19 | 118 |
| 20 | 117 |
| 21 | 123 |
| 22 | 114 |
| 23 | 119 |
| 24 | 120 |
| 25 | 118 |

Perform a one-sample z-test at a significance level of $\alpha = 0.05$ to decide whether to reject or fail to reject the null hypothesis based on the p-value and significance level.

## Problem 1: One-Sample Z-Test for Blood Pressure Data

### Step 1: Calculate the Sample Mean and Standard Deviation

Given the blood pressure data, we calculate:

$$\bar{x} = \frac{\sum_{i=1}^{25} x_i}{25} = 118.48\,\text{mmHg}$$

The sample standard deviation is:

$$s = \sqrt{\frac{\sum_{i=1}^{25} (x_i - \bar{x})^2}{24}} = 2.6633\,\text{mmHg}$$

### Step 2: Formulate Hypotheses

- Null Hypothesis ($H_0$): $\mu = \mu_0 = 120$ mmHg
- Alternative Hypothesis ($H_a$): $\mu \neq \mu_0$

### Step 3: Calculate the Z-statistic

The z-statistic is calculated using the formula:

$$z_{\text{stat}} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Where:

$$\begin{aligned}
\bar{x} &= 118.48\,\text{mmHg}\,\text{(sample mean)} \\
\mu_0 &= 120\,\text{mmHg}\,\text{(population mean)} \\
\sigma &= 10\,\text{mmHg}\,\text{(population standard deviation)} \\
n &= 25\,\text{(sample size)}
\end{aligned}$$

Substituting these values into the formula gives:

$$z_{\text{stat}} = \frac{118.48 - 120}{10/\sqrt{25}} = -0.76$$

### Step 4: Determine the p-value

The p-value is determined from the z-statistic using a standard normal distribution. Since this is a two-tailed test (as we are testing for any difference, not just an increase or decrease), the p-value is calculated as:

$$p\text{-value} = 2 \times (1 - \Phi(|z_{\text{stat}}|))$$

Calculating from z-table, for our z-statistic of $z_{\text{stat}} = -0.76$, the p-value is approximately:

$$p\text{-value} = 0.4472$$

where $\Phi$ is the cumulative distribution function of the standard normal distribution.

## Step 5: Decision

At a significance level of $\alpha = 0.05$, since $0.4472 > 0.05$, we fail to reject the null hypothesis.

Thus, there is not enough evidence to conclude that the new diet affects blood pressure levels in this sample.



Reference: Steps to calculate the p-value for a z-score of -0.76 using a standard normal distribution table, follow these steps:

1. **Locate the Z-Score in the Table:** Standard normal distribution tables typically provide the cumulative probability (area under the curve to the left) for positive z-scores. For a z-score of -0.76, you would first find the cumulative probability for 0.76 (since the distribution is symmetric).
2. **Find the Cumulative Probability:** Look up 0.76 in the z-table. The table will provide the cumulative probability for this z-score. Typically, this value is around 0.7764.
3. **Calculate the P-Value for Two-Tailed Test:**
   - Since you are dealing with a two-tailed test (testing for any difference), you need to consider both tails of the distribution.
   - The cumulative probability for -0.76 is calculated as:
     $$1 - 0.7764 = 0.2236$$
   - For a two-tailed test, multiply this by 2 to account for both tails:
     $$2 \times 0.2236 = 0.4472$$

Thus, the p-value for a z-score of -0.76 in a two-tailed test is approximately 0.4472.